

# From Probability to Optimization: The Foundations of Machine Learning

IOAI Preparation

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## Introduction

Machine Learning (ML) can be understood through two complementary perspectives:

- **Statistical View (Maximum Likelihood):** We assume that the data is generated by an underlying probabilistic model (for instance, a Gaussian distribution with parameters  $\mu, \sigma$ ), and we estimate the parameters that make the observed data most likely. This is done by maximizing what we call the likelihood of the data.
- **Optimization View (Empirical Risk Minimization):** We define a loss function  $\ell$  that measures prediction error, and we minimize the average loss over the dataset using tools from calculus.

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This problem set is to understand both methods and give out a few examples of their usage.

0) Give examples of models you know for each of these two approaches.

## 1 Optimization View

### 1.1 Probabilistic Foundation: Why Learning Works

To justify using a finite dataset to approximate reality, we rely on the **Law of Large Numbers**. For an event  $A$ , we define the indicator random variable  $\mathbf{1}_A$  by:

$$\mathbf{1}_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

1. Show that  $\mathbb{E}(\mathbf{1}_A) = \mathbb{P}(A)$ .
2. **Markov's Inequality:** Let  $Y$  be a non-negative random variable and  $a \geq 0$  be a real number.
  - (a) Show that  $a \cdot \mathbf{1}_{Y \geq a} \leq Y$ . **Hint:** Prove the inequality first when  $Y \geq a$  and then when  $Y < a$ .

(b) Deduce that:

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

3. **Bienaymé-Tchebychev Inequality:** Apply Markov's inequality to  $Y = (X - \mathbb{E}[X])^2$  to prove:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

4. **Empirical Mean:** Let  $X_1, \dots, X_n$  be independent and identically distributed (i.i.d) random variables with mean  $\mu$  and variance  $\sigma^2$ , and let  $\bar{X}_n$  be their average. In other words  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

(a) Show that  $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$ . You may start by computing  $V(X + Y)$  when  $X, Y$  are independent and  $V(aX)$  when  $a \in \mathbb{R}$ . Recall that if  $X$  and  $Y$  are independent, then  $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ .

(b) Deduce that for any  $\varepsilon > 0$ ,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) = 0.$$

This is known as the weak law of large numbers. Explain what this means.

**Remark (Strong Law of Large Numbers).** The theorem we just proved is the weak law of large numbers. In fact, the following stronger statement holds: If  $X_1, X_2, \dots$  are i.i.d. random variables with finite mean  $\mu$ , then:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

## 5. Conclusion (Learning Principle):

We are given data  $(X_i, Y_i)_{1 \leq i \leq N}$  drawn from an unknown distribution. We consider a model  $f_\theta$  and a loss function  $\ell$ .

- (a) Give  $f_\theta$  in the case of linear regression. In the case of a neural network, what does  $f_\theta$  look like ?
- (b) In the general case, why is it natural to minimize the empirical loss:

$$\frac{1}{N} \sum_{i=1}^N \ell(f_\theta(X_i), Y_i).$$

You may use the law of large numbers for this.

## 1.2 Geometric Foundation: Cauchy-Schwarz Inequality

We now understand **what** we need to optimize and **why** (if it is still unclear, go back to the previous problem). However, before optimizing, we must understand geometry in high-dimensional spaces. In particular, we prove the following fundamental result which will help us later:

**Cauchy-Schwarz Inequality:** For any vectors  $u, v \in \mathbb{R}^d$ ,

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

For this, we consider  $P(t) = \|u + tv\|^2$  for  $t \in \mathbb{R}$ .

1. Let  $a, b \in \mathbb{R}^n$ . Show that

$$\|a + b\|^2 = \|a\|^2 + \|b\|^2 + 2\langle a, b \rangle.$$

2. Expand  $P(t)$  to show that it is a quadratic polynomial of the form  $At^2 + Bt + C$ . Identify  $A, B$ , and  $C$ .
3. What is the sign of  $P(t)$ ? Deduce the sign of  $\Delta = B^2 - 4AC$ .
4. Conclude the proof of the Cauchy-Schwarz inequality. In the case of  $\mathbb{R}^2$ , can you come up with a direct proof?

### 1.3 Optimization: Gradient Descent

We now have the tools to understand the main optimization algorithm: gradient descent. The idea is simple: to minimize a function  $f(x)$ , we move in the direction of steepest descent. We start by showing how to do that and then prove a guarantee on gradient descent.

1. Using the approximation

$$f(x + h) \approx f(x) + \langle \nabla f(x), h \rangle,$$

and the Cauchy-Schwarz inequality, show that for a fixed step size  $\|h\| = \alpha$ , the quantity  $\langle \nabla f(x), h \rangle$  is minimized when  $h$  is in the opposite direction of the gradient.

2. Write the recurrence relation which defines gradient descent on model weights  $(\theta_k)_{k \geq 1}$ . Explain it using the previous question. You may draw a graph in 3D to explain.
3. **Convergence:** Suppose  $f$  is  $L$ -smooth i.e. suppose that:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

This may seem arbitrary but many loss functions (like the linear regression least squares loss) are  $L$ -smooth for some  $L$ .

If we define

$$\theta_{k+1} = \theta_k - \frac{1}{L} \nabla f(\theta_k),$$

First explain what role  $\frac{1}{L}$  plays here, then show that:

$$f(\theta_{k+1}) \leq f(\theta_k) - \frac{1}{2L} \|\nabla f(\theta_k)\|^2.$$

Conclude that, if  $f$  has a minimum, then  $(f(\theta_k))_{k \geq 0}$  converges.

4. Denote by  $f_*$  the minimum of  $f$ . Show that for  $T > 0$ :

$$\min_{0 \leq t \leq T-1} \|\nabla f(\theta_t)\|^2 \leq \frac{2L(f(\theta_0) - f_*)}{T}$$

Explain what this inequality means.

**Remark:** It would be possible to go further and obtain the speed of convergence of  $(\theta_k)$  to a minimum of  $f$ . Under some extra assumptions on  $f$  (namely strong convexity), it can be shown that the convergence is at least geometric.

5. Alice & Bob are training a neural network on a large dataset.

Alice argues they should use full gradient descent: “our convergence proof shows that gradient descent strictly decreases the loss at every step, so we will find a more precise minimum.”

Bob disagrees and wants to use stochastic gradient descent (SGD), where at each step you only compute the gradient on a small random batch of the data instead of the full dataset.

- (a) Alice claims SGD is less precise than gradient descent because the gradient estimate is noisy. Is she right? What does “precise” even mean in this context — should we care about minimizing the training loss as accurately as possible? Think about it using question 4.a. of problem 1.
- (b) Bob points out a practical issue with full gradient descent when the dataset has one million points. What is it?
- (c) In practice, models trained with SGD often *generalize better* to new data than those trained with full gradient descent, even though SGD is noisier. Can you think of an intuitive reason why noise during training might actually be beneficial?
- (d) Conclude: who is right, Alice or Bob? Or is the question itself flawed?

## 2 Statistical View

### Transition: The Concept of Likelihood

In Machine Learning, we often observe a dataset  $\mathcal{D} = \{x_1, x_2, \dots, x_n\}$  following a distribution with parameters  $\theta$ , and must infer the unknown parameters  $\theta$  that generated it.

To bridge this gap, we introduce the **Likelihood function**, denoted by  $L(\theta)$ . Unlike a probability distribution which varies over the data  $x$  for a fixed  $\theta$ , the likelihood function varies over the parameter space  $\theta$  for the fixed, observed data  $\mathcal{D}$ .

**Formal Definition** Let  $X_1, \dots, X_n$  be i.i.d. random variables with a probability density (or mass) function  $f(x; \theta)$ . Given the observed values  $x_1, \dots, x_n$ , the likelihood of  $\theta$  is defined as the joint probability of the data:

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta)$$

**The Principle of Maximum Likelihood (MLE)** The “most likely” value for the parameter  $\theta$  is the one that maximizes the probability of having observed our specific dataset (read this sentence as many times as you need to understand it: it is the core of this topic). We define the Maximum Likelihood Estimator (MLE) as:

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

## Intuitive Examples

- **The Coin Flip:** If you flip a coin 10 times and get 9 Heads, you naturally assume the coin is biased toward Heads. Mathematically, you are saying that  $\mathbb{P}(\text{Heads}) = 0.9$  makes the observed sequence much more *likely* than  $\mathbb{P}(\text{Heads}) = 0.5$ .
- **The Height Model:** If you measure three students at 180cm, 182cm, and 179cm, a model with a mean  $\mu = 160\text{cm}$  would give a very low "score" (likelihood) to these points. A model with  $\mu = 180\text{cm}$  yields a much higher likelihood because it places the highest probability density where the data actually sits.

In the following subsection, we will see that maximizing this product is often easier by maximizing its logarithm,  $\ln(L(\theta))$ , known as the **log-likelihood**.

**Worked Example: Estimating a Coin's Bias** Suppose we flip a coin  $n$  times and observe  $k$  heads. The coin lands heads with unknown probability  $\theta \in [0, 1]$ . Each flip is independent, so the likelihood of our dataset is:

$$L(\theta) = \theta^k (1 - \theta)^{n-k}.$$

Rather than maximizing  $L(\theta)$  directly, we maximize the **log-likelihood**  $\ell(\theta) = \ln L(\theta)$ , which is equivalent (since  $\ln$  is strictly increasing) but much easier to work with:

$$\ell(\theta) = k \ln \theta + (n - k) \ln(1 - \theta).$$

Setting the derivative to zero:

$$\frac{d\ell}{d\theta} = \frac{k}{\theta} - \frac{n-k}{1-\theta} = 0 \implies k(1-\theta) = (n-k)\theta \implies \hat{\theta}_{\text{MLE}} = \frac{k}{n}.$$

**Interpretation:** The MLE simply says "*the most likely parameter is the one that matches what we observed.*" If we flipped 10 times and got 7 heads, our best guess is  $\hat{\theta} = 0.7$ . This aligns perfectly with intuition, and as  $n \rightarrow \infty$ , the Law of Large Numbers guarantees that  $\hat{\theta}_{\text{MLE}} \rightarrow \theta_{\text{true}}$ .

## 2.1 Warm-up: Understanding Maximum Likelihood

1. A biased coin lands heads with probability  $p$ . You observe the sequence:

$$H, H, T, H.$$

- (a) Write the likelihood  $L(p)$  of observing this sequence.
  - (b) Find the value of  $p$  that maximizes  $L(p)$ . We will denote it as  $\hat{p}$ . **Hint:** finding  $p$  which maximizes  $L(p)$  is the same as finding  $p$  which maximizes  $\ln(L(p))$
  - (c) Generalize to the case where you observe a sequence  $X_1, \dots, X_n$ . Does the result make sense to you? What can you say using the law of large numbers? What does that entail if you throw the coin enough times?
2. Let  $X_1, \dots, X_n$  be generated with distribution  $\mathcal{N}(\mu, 1)$ .
    - (a) Give an example of a real dataset that might follow this distribution.
    - (b) Write the likelihood of the dataset. **Hint:** In the continuous case, the likelihood is the product of the densities evaluated in each  $X_i$ .

- (c) Show that the maximum likelihood estimator of  $\mu$  is the empirical mean. Does the result make sense to you ? What can you say using the law of large numbers ? What does that entail if you have a lot of data ?
- 3. In your own words, explain what it means to “maximize the likelihood”. Explain the logistic regression using this concept.

## 2.2 The Synthesis: Linear Regression, MLE, and Model Assumptions

We now revisit linear regression from both a statistical and optimization perspective.

1. **Model Definition:** Let  $(x_i, y_i)_{1 \leq i \leq n}$  with  $x_i \in \mathbb{R}^d$ . Write the form of a linear model  $f_\theta(x)$  for linear regression.
2. **Least Squares:** We estimate  $\theta$  by minimizing:

$$\frac{1}{n} \sum_{i=1}^n (f_\theta(x_i) - y_i)^2.$$

Explain why this is an optimization problem. Can you solve it analytically ?

3. **Statistical Interpretation (MLE):** Assume:

$$Y_i = \langle \theta, X_i \rangle + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- (a) Write the likelihood of the dataset  $(Y_i)_{1 \leq i \leq N}$ . **Note:** This is a statistical analysis where the  $X_i$  are seen as fixed real numbers, not random variables. Start by explaining why  $Y_i \sim \mathcal{N}(\langle \theta, X_i \rangle, \sigma^2)$
  - (b) Show that maximizing the likelihood is equivalent to minimizing the squared loss.
4. **Discussion: Is Linear Regression an MLE problem?**
    - (a) Under which assumptions does linear regression correspond to MLE?
    - (b) What happens if the noise is not Gaussian?

## 2.3 Final Challenge: Alice & Bob’s Secret Machine

Bob has built a mysterious machine.

Each time he presses one of three hidden buttons, the machine produces a point in the plane  $(x, y)$ . More precisely, with probability  $\pi_1$ , he picks the red button, with probability  $\pi_2$  the green one and with probability  $\pi_3$  the blue one. When Bob generates a point after choosing a button, the color of the point is the color of the chosen button.

Alice is given a dataset of points  $(x_i, y_i)$  together with the color of the button that was pressed.

Bob now gives Alice a new dataset of points, but this time he does *not* reveal the colors. The datasets are *train.csv* (labeled) and *test.csv* (unlabeled).

**Your goal is to understand how Bob’s machine works and predict the missing colors.**

1. **A first approach (no probability).**

Using two different models of your choice that do not use maximum likelihood estimation, propose a method to assign a color (red, blue, or green) to a new point  $(x, y)$ .

- (a) Explain your methods clearly. Explain why they are appropriate for this situation.
- (b) Apply them to the provided test dataset.

## 2. A probabilistic model.

By looking at the shape of the data, propose a probabilistic model that could explain how Bob generates the points. More precisely, give a distribution which seems to fit the data properly. Clearly describe the parameters of your model.

## 3. Maximum Likelihood. (This problem is on the harder end)

Suppose your model depends on parameters  $\theta$ . Write the likelihood of the dataset under your model.

If you had access to the parameters maximizing this likelihood, explain how you would assign a color to a new unlabeled point  $(x, y)$ . More generally, explain how your probabilistic model naturally leads to a method for clustering the data points.

## 4. Bonus (This is a hard problem : think of it as a research task).

We wish to optimize the likelihood you wrote. Suggest a strategy to estimate the parameters of your model.

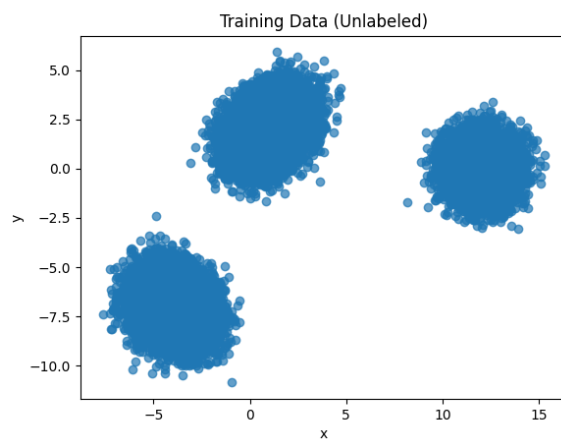
**Remark:** The following images are not exactly the images of the data you have been given. Make sure to plot your data !

## 2.4 Conclusion

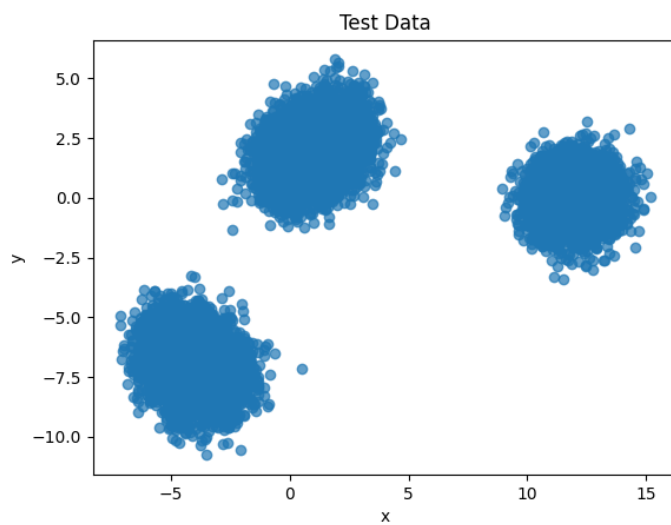
Make a report of the main takeaways for these problems. You don't need to make a good looking report, just make sure you write down what you understood somewhere.



(a) Training Data (Labeled)



(b) Training Data (Unlabeled)



(c) Test Data

Figure 1: Visualizations of the data.